Muhammad Talha Atif

Software Engineer

+923064391520 ◇ talhaatif573@gmail.com ◇ Lahore, Pakistan ◇ LinkedIn ◇ GitHub ◇ Portfolio

SUMMARY

Software Engineer with 2+ years of production experience building API-driven backend systems, third-party integrations, and scalable web applications. Designed and shipped REST APIs with reliability patterns including rate limiting, retry with exponential backoff, dead-letter queues, and idempotent operations across distributed microservices. Implemented OAuth 2.0, JWT-based authentication, and secure credential handling in production systems. Built lifecycle state machines orchestrating entity transitions across multi-service workflows. Comfortable with AI-assisted developer tooling for accelerating delivery. Strong fundamentals in data modelling, system design, and internal-to-external state mapping.

WORK EXPERIENCE

Software Engineer *Dec '25 – Jun '26*
Intimec via MicroTechX Remote

Built and owned API-driven backend services and integration infrastructure for an enterprise Identity Governance platform, designing lifecycle state machines, reliability patterns, and cross-service orchestration across 8+ distributed microservices.

- Designed and implemented the access request lifecycle state machine orchestrating entity transitions across submission, validation, approval, provisioning, and rollback states, with idempotent operations ensuring safe retries and consistent state across distributed services.
- Built Kafka-based integration pipelines with rate-limit-aware consumer concurrency, exponential backoff retry windows, dead-letter queues for partial failure recovery, and correlation-ID-based idempotent consumers preventing duplicate processing.
- Migrated inter-service communication from REST to gRPC with Protocol Buffers, reducing integration latency by 50% across Auth, Identity, and Access Portal services, and maintained internal-to-external ID mapping across service boundaries.
- Implemented Spring Security with OAuth 2.0 token validation, JWT-based session management, RBAC enforcement, and secure credential handling with scoped permissions across all API endpoints.
- Authored internal documentation covering local environment setup, service boundaries, API contracts, debugging guides, and seed test data, eliminating onboarding friction for new engineers joining a previously undocumented codebase.

Software Engineer *Sep '25 – Jan '26*
Digital Upgraders LLC Remote

Built and maintained API-driven backend services and React frontend features for a production social platform, implementing reliability patterns, rate limiting, and performance optimisations across a system serving real users daily.

- Cut feed API response time from 3 seconds to 700ms by profiling query execution, adding composite indexes on read-heavy columns, pruning unused DTO fields, and implementing a TTL-based in-memory cache with pagination support.
- Implemented IP-based rate limiting across authentication and high-frequency API endpoints, and designed transactional write patterns replacing non-atomic queries to eliminate notification duplication under concurrent load.
- Led zero-downtime production migration covering Docker containerisation, DNS cutover, CI/CD pipeline reconfiguration, and environment management across a live multi-service platform with no user-facing disruption.

PROJECTS

Moodly | AI Experience Booking Platform | Demo

- Built RESTful APIs for the full booking lifecycle with input validation, JWT authentication, and an async RabbitMQ pipeline with retry logic decoupling AI inference from the booking engine, handling partial failures gracefully.
- Integrated third-party AI model APIs (Hugging Face, Gemini) in production with structured error handling, request timeout management, and fallback responses, translating partner validation errors into actionable user-facing messages.

Memingo | Language Learning Platform | Demo

- Resolved a leaderboard bottleneck by replacing per-request computation with precomputed Redis cache, reducing API response from 30 seconds to 90ms.
- Integrated Gemini API with session-aware context management, structured output parsing, and retry logic for transient API failures.

SKILLS

Backend Java, Spring Boot, Spring Security, Node.js, Express.js, NestJS, FastAPI, Python, REST APIs, gRPC, GraphQL (familiar)

Frontend React.js, Next.js, React Native, TypeScript, JavaScript, Tailwind CSS

Integration Third-party API integration, rate limiting, retry/backoff, pagination, idempotent operations, circuit breaker patterns

Auth OAuth 2.0, JWT, RBAC, secure credential handling, token refresh, scoped permissions

Architecture Microservices, event-driven architecture, lifecycle state machines, Kafka, RabbitMQ, WebSockets

Databases PostgreSQL, MySQL, MongoDB, Redis, OpenSearch, Supabase, Firebase

Cloud / DevOps AWS (S3, RDS, EC2), Docker, GitHub Actions, CI/CD, Fly.io, Dokploy

AI Tooling GitHub Copilot, Claude AI, Gemini API, Hugging Face, LangChain (familiar)

OPEN SOURCE

Apache Checkstyle Merged PR #18807
Contributed to Apache-maintained static analysis tool used across thousands of production Java projects globally.

AWARDS

Runner-up, App Development Hackathon 2025 *May '25*
Punjab University

Ranked, ICPC Asia Topi Online Programming Contest 2024 *Nov '24*
International Collegiate Programming Contests

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology, UMT *Oct '21 – Aug '25*